

When Standard Methods Succeed

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when correlation *is* causation

When you have no confounders and there is a linear relationship between the exposure and the outcome, that *correlation is a causal relationship*



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randomized controlled trials

A/B testing

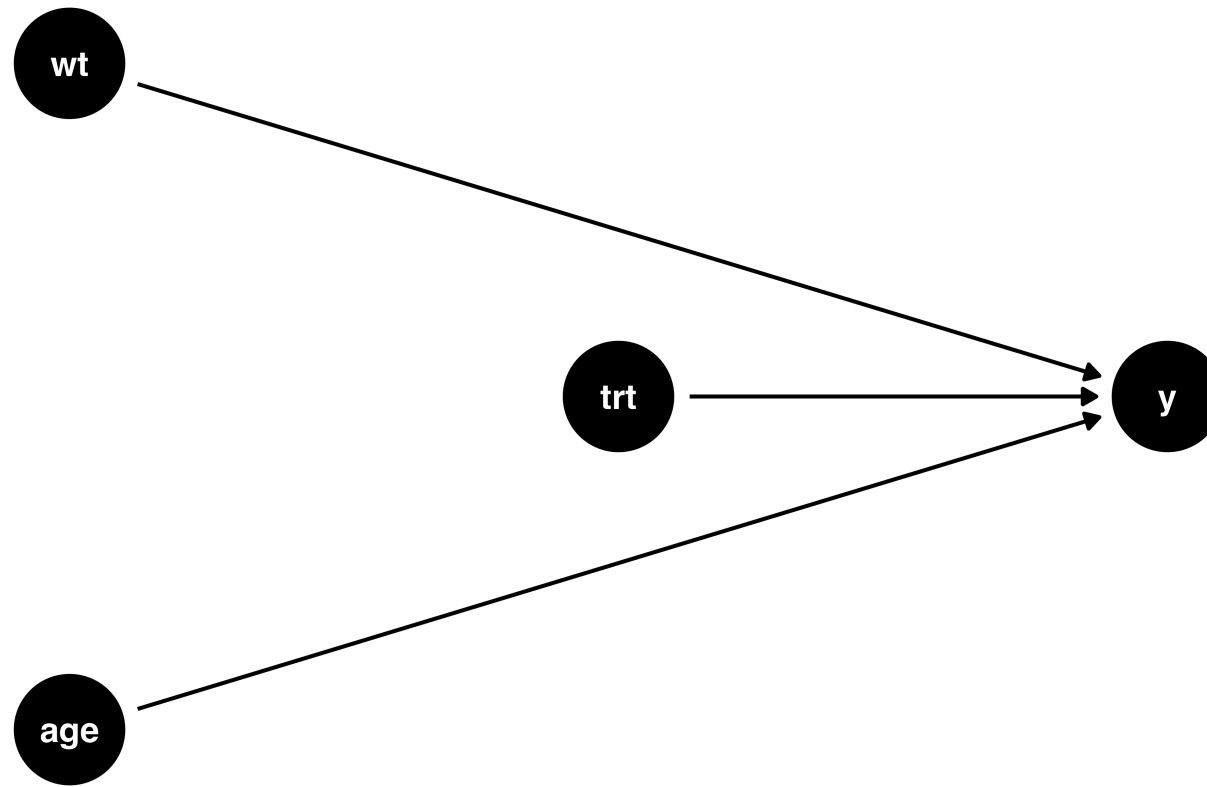
Even in these cases, using the methods you will learn here can help!

- 1 Adjusting for baseline covariates can make an estimate *more efficient*
- 2 Propensity score weighting *always improves precision* over no adjustment and is *equivalent to direct adjustment*
- 3 Sometimes we are *more comfortable with the functional form of the propensity score* (predicting exposure) than the outcome model

Example

- simulated data (100 observations)
- Treatment is randomly assigned
- There are two baseline covariates: age and weight

Example

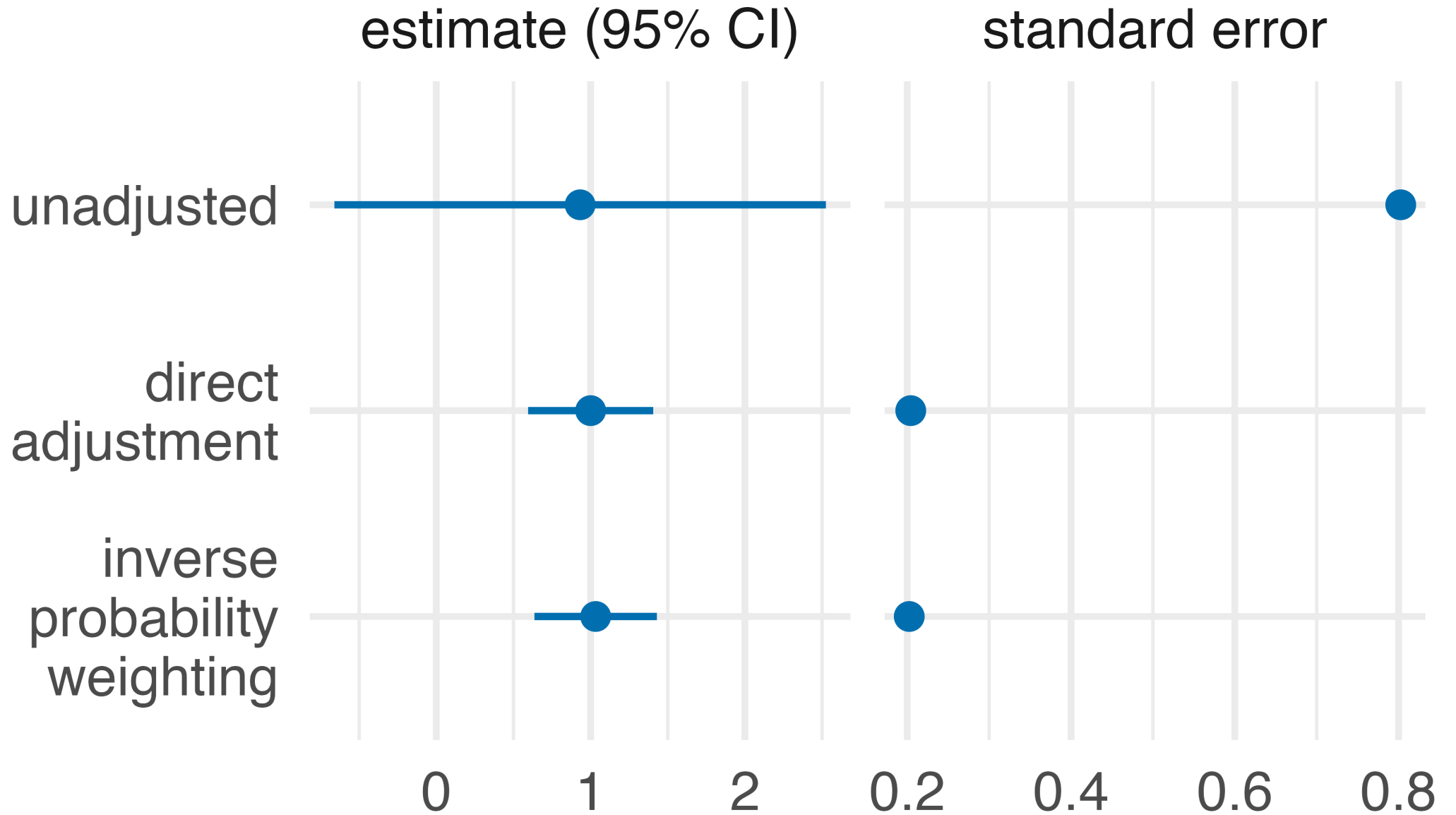


- True average treatment effect: 1

Three ways to estimate the effect

```
1 unadjusted <- lm(y ~ treatment, data = d)
2
3 adjusted <- lm(y ~ treatment + weight + age, data = d)
4
5 propensity_model <- glm(
6   treatment ~ weight + age,
7   family = binomial(),
8   data = d
9 )
10 d_wts <- propensity_model |>
11   augment(data = d, type.predict = "response") |>
12   mutate(wts = wt_ate(.fitted, treatment, exposure_type = "binary"))
13 outcome_model <- lm(y ~ treatment, data = d_wts, weights = wts)
14 weighted <- ipw(propensity_model, outcome_model)
```

`ipw()` uses a sandwich estimator that also accounts for the uncertainty in the propensity score model

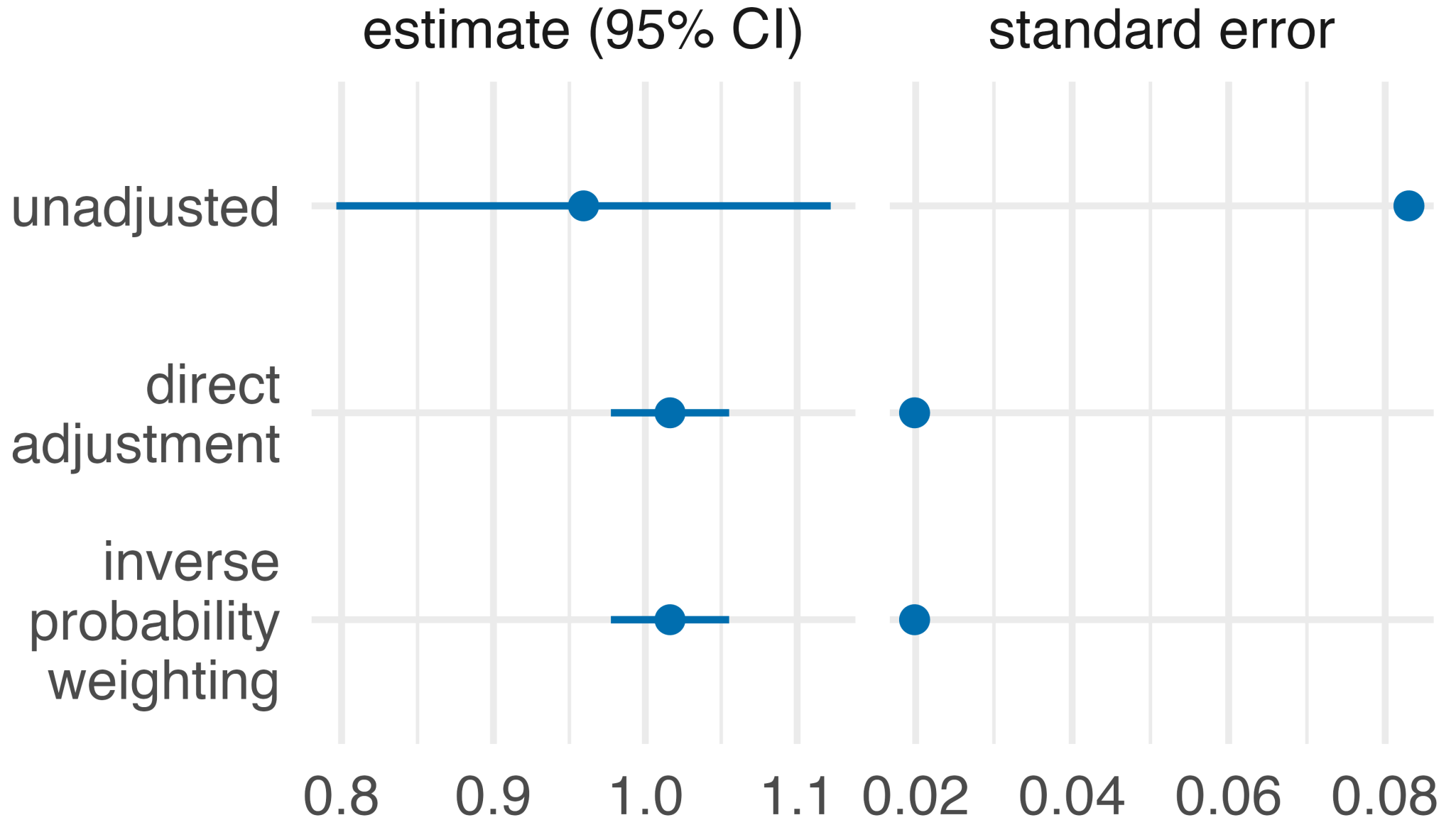


Conditional vs. marginal effects

- The outcome model gives a **conditional** effect: *holding the other variables constant*
- Weighting gives a **marginal** effect: *averaged over the population*
- With a continuous outcome, a linear effect, and no interactions, the two are identical

Example

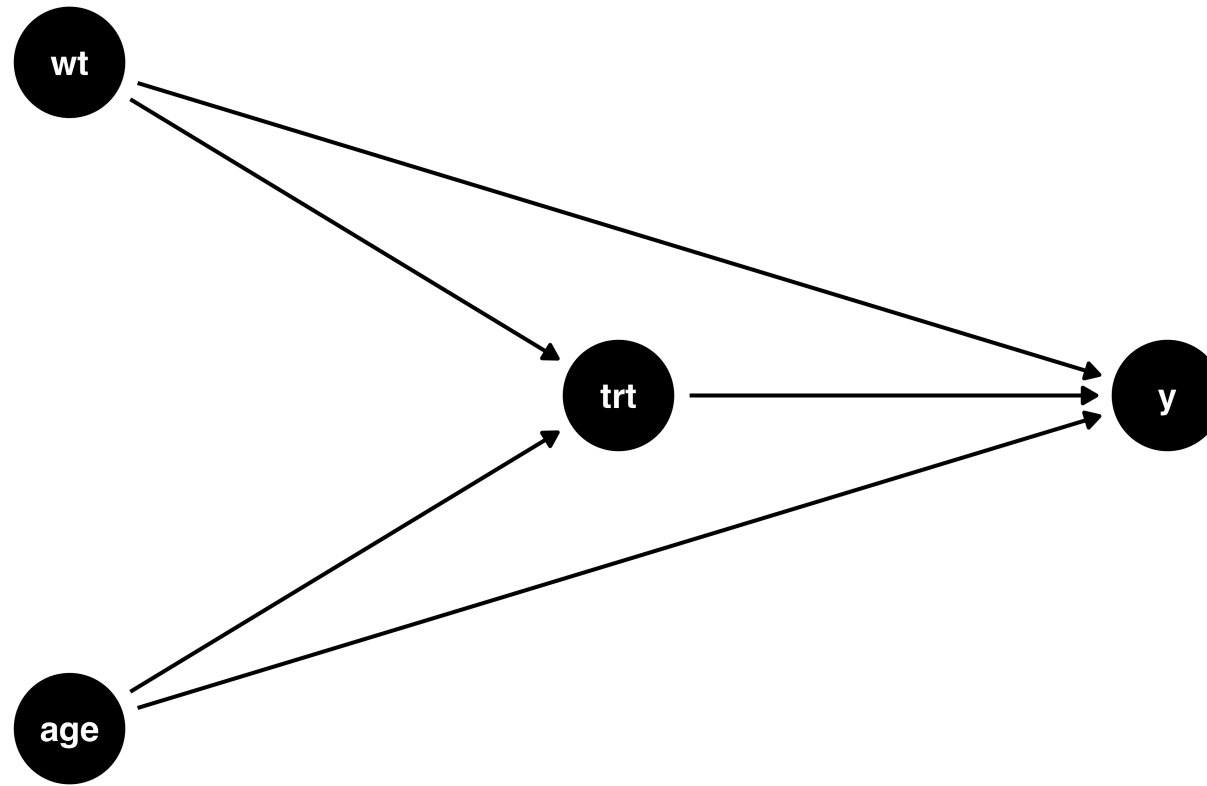
- simulated data (10,000 observations)
- Treatment is randomly assigned
- There are two baseline covariates: age and weight



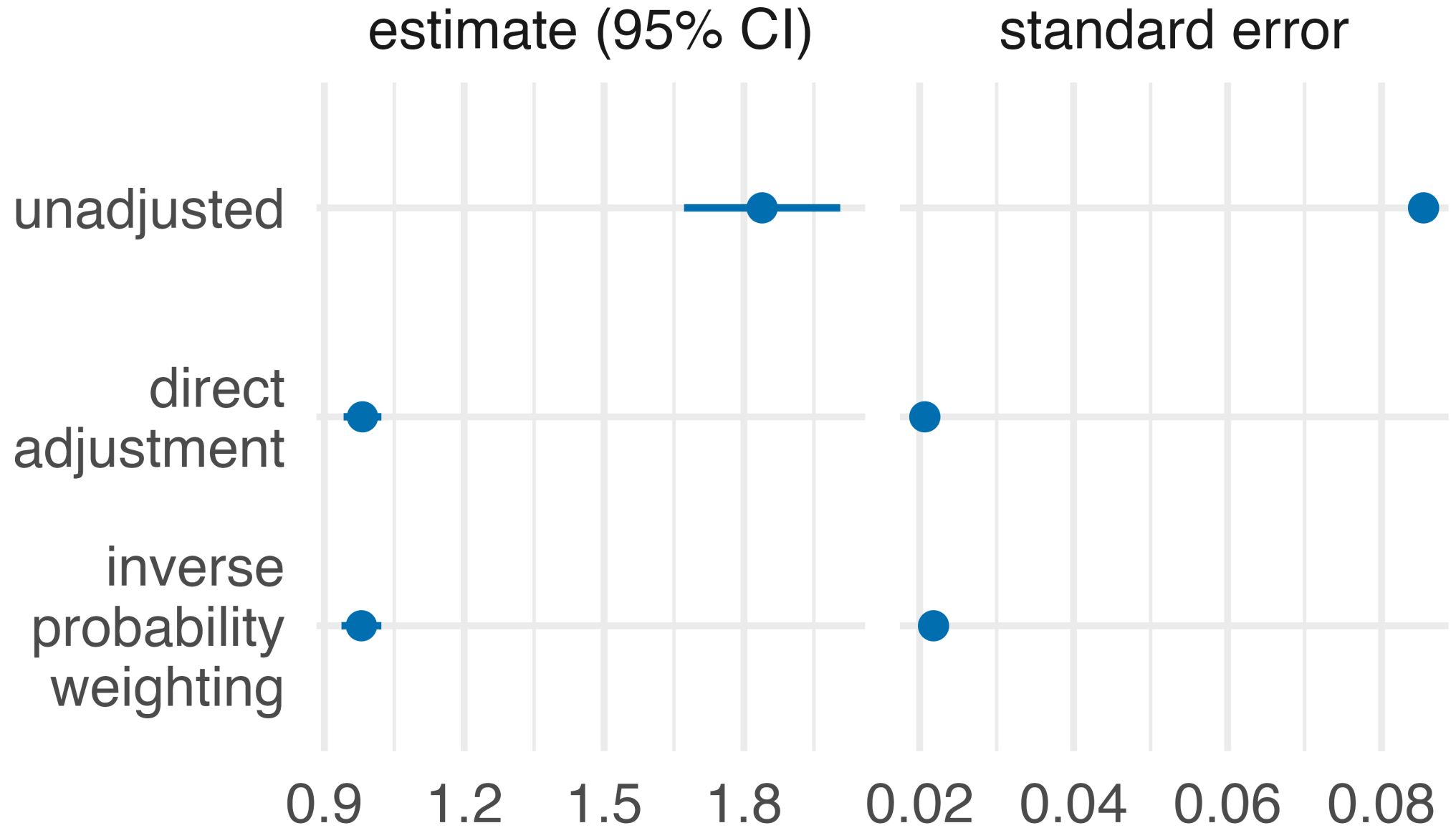
Example

- simulated data (10,000 observations)
- Treatment is **not** randomly assigned
- Two baseline confounders: age and weight
- The effect is **homogeneous**, so marginal and conditional effects agree

Example



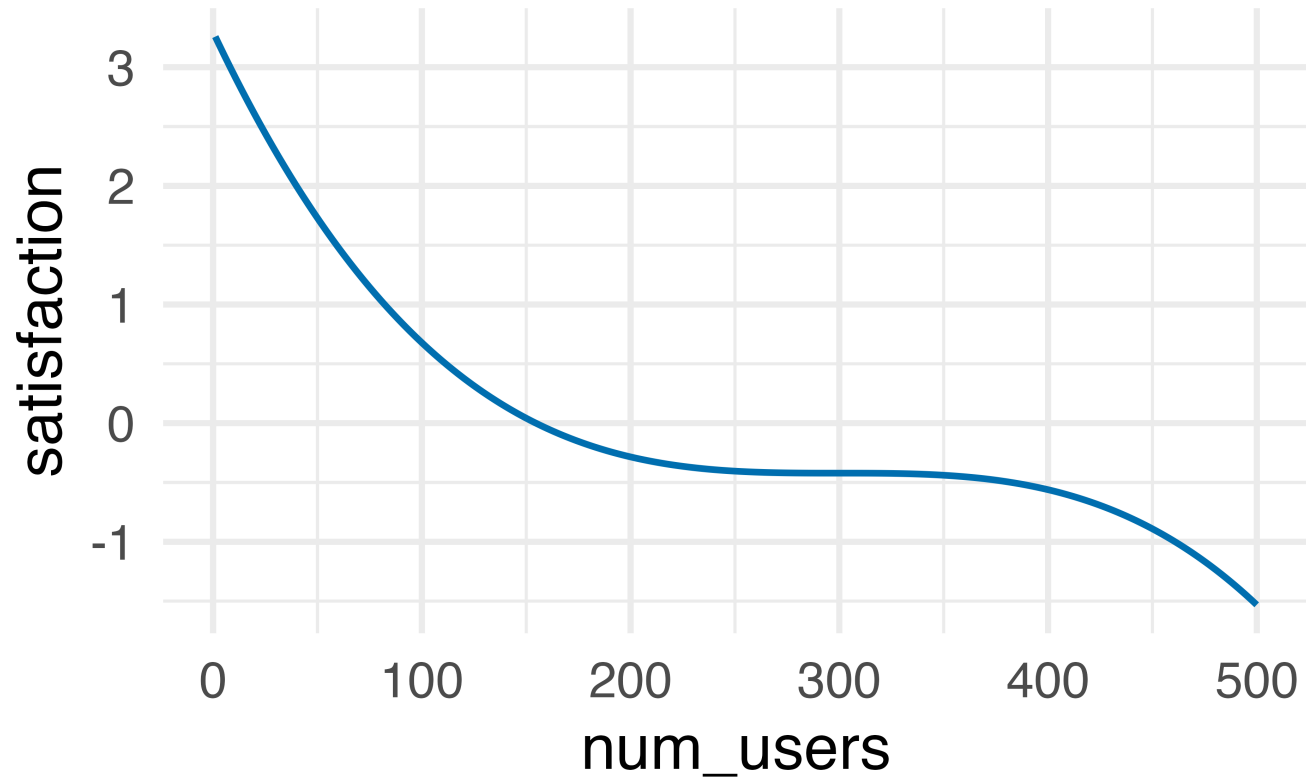
- True average treatment effect: 1



Example

- simulated data (10,000 observations)
- The exposure is **daily updates**, the outcome is **satisfaction**
- There is one baseline confounder: `num_users`
- True average treatment effect: 0

Example



- The relationship is **non-linear**

The wrong functional form

```
1 lm(satisfaction ~ update_frequency + num_users, data = d) |>  
2   tidy(conf.int = TRUE) |>  
3   filter(term == "update_frequencydaily") |>  
4   select(estimate, starts_with("conf"))
```

```
# A tibble: 1 × 3  
  estimate conf.low conf.high  
  <dbl>    <dbl>    <dbl>  
1  -0.189  -0.219    -0.159
```

- The estimate is far from the truth, which is not even contained in the confidence interval

A flexible functional form

```
1 lm(  
2   satisfaction ~ update_frequency + splines::ns(num_users, 3),  
3   data = d  
4 ) |>  
5 tidy(conf.int = TRUE) |>  
6 filter(term == "update_frequencydaily") |>  
7 select(estimate, starts_with("conf"))
```

```
# A tibble: 1 × 3  
  estimate conf.low conf.high  
    <dbl>    <dbl>    <dbl>  
1  0.00216 -0.00258  0.00690
```

- We often do not know the data-generating mechanism, but we can still fit flexible parametric models

**the model needs the right *variables* and the right
*functional form***

We need the correct functional form for both the exposure and the
confounders

time-varying **confounding**

Linear regression cannot estimate these types of effects without bias, so we'll need methods like inverse probability weighting or g-computation to calculate it correctly